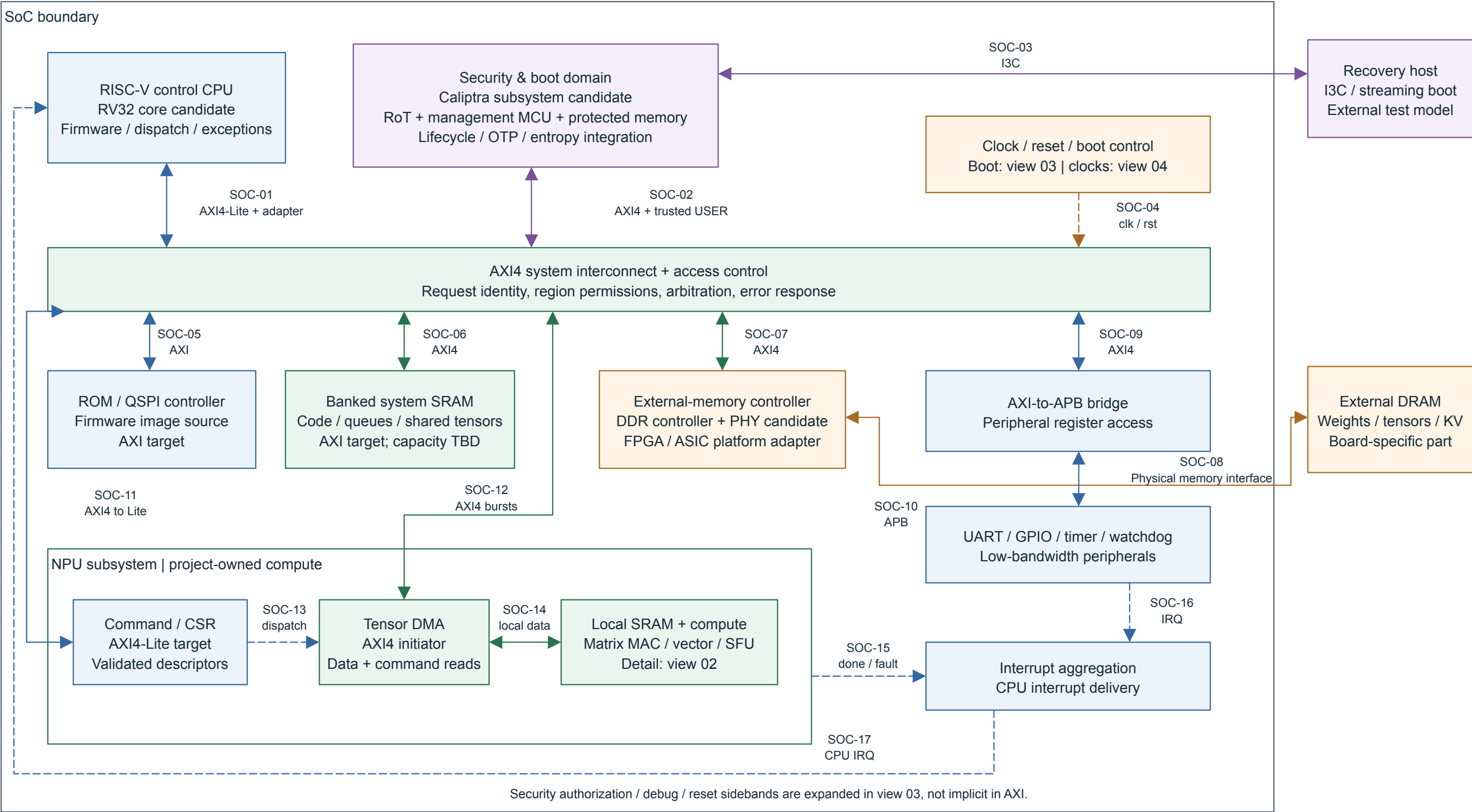


OpenChip NPU SoC | Top-level block diagram

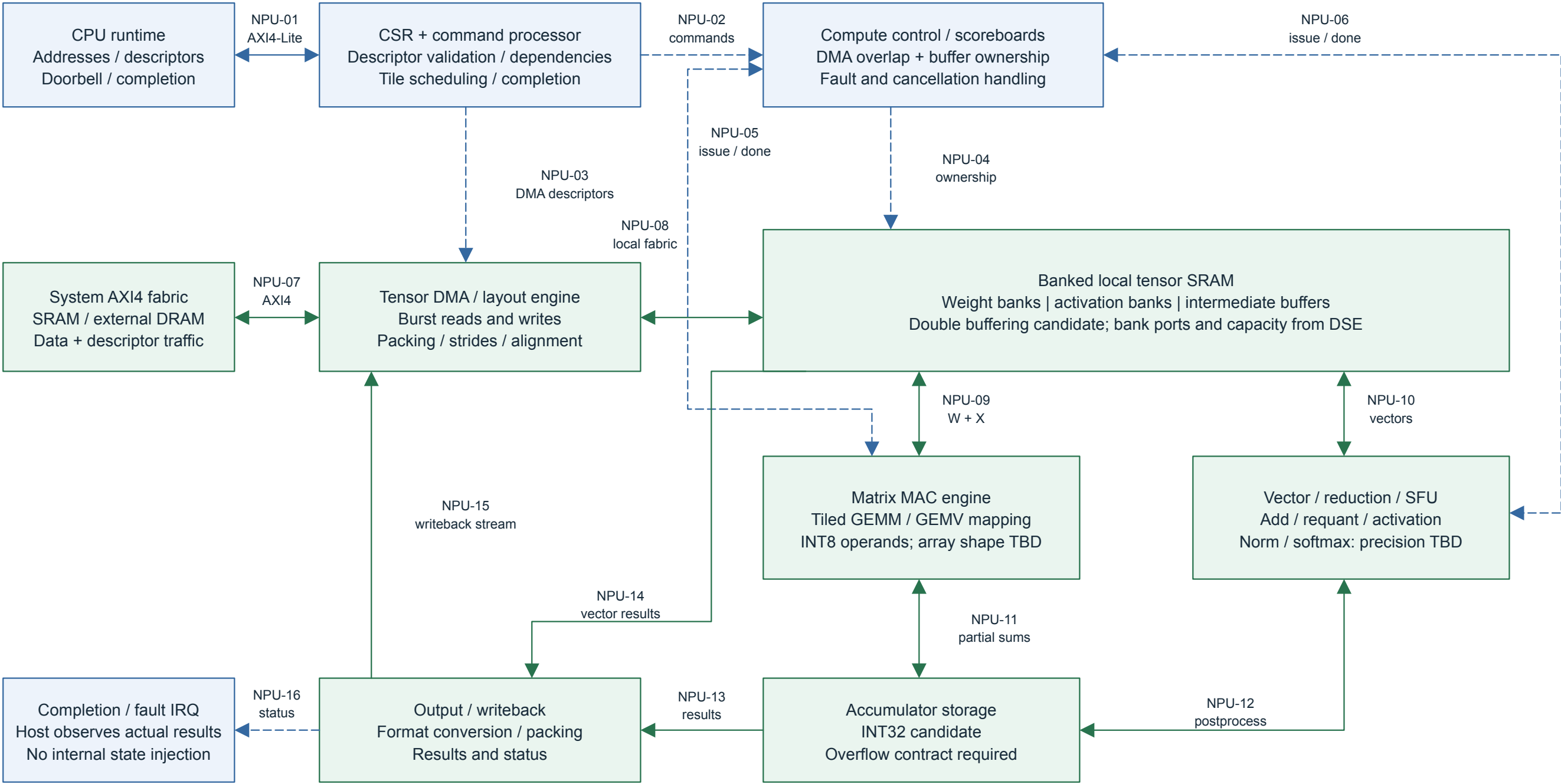
OpenChip architecture review | Rev 0.2 | Architecture candidate v0.2 | logical connectivity, not floorplan | LOCAL DRAFT



Solid arrows: interface traffic (double = bidirectional). Dashed: control / IRQ. Crossings without dots are not junctions.

OpenChip NPU | Compute and data movement

OpenChip architecture review | Rev 0.2 | Functional organization | candidate INT8 MAC + INT32 accumulation | precision / sizing not frozen

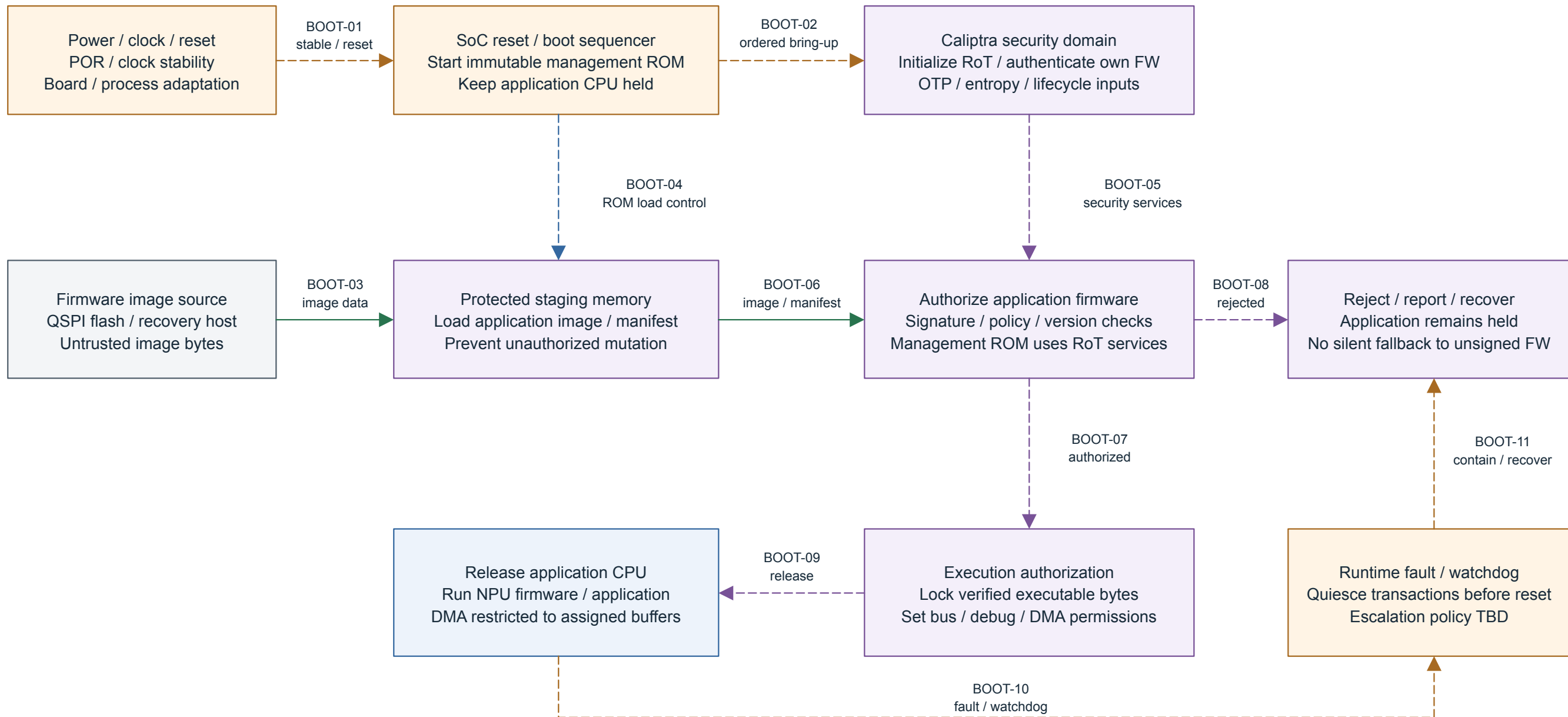


Local data paths are SRAM/native or explicitly adapted streams; ready/valid alone does not claim AXI4-Stream compliance.

GEMV utilization and non-MAC operators need separate measurement; theoretical MAC count is not application throughput.

OpenChip | Reset, secure boot and execution authorization

OpenChip architecture review | Rev 0.2 | Proposed security-enabled system flow | generic roles, not a Caliptra signal-level specification

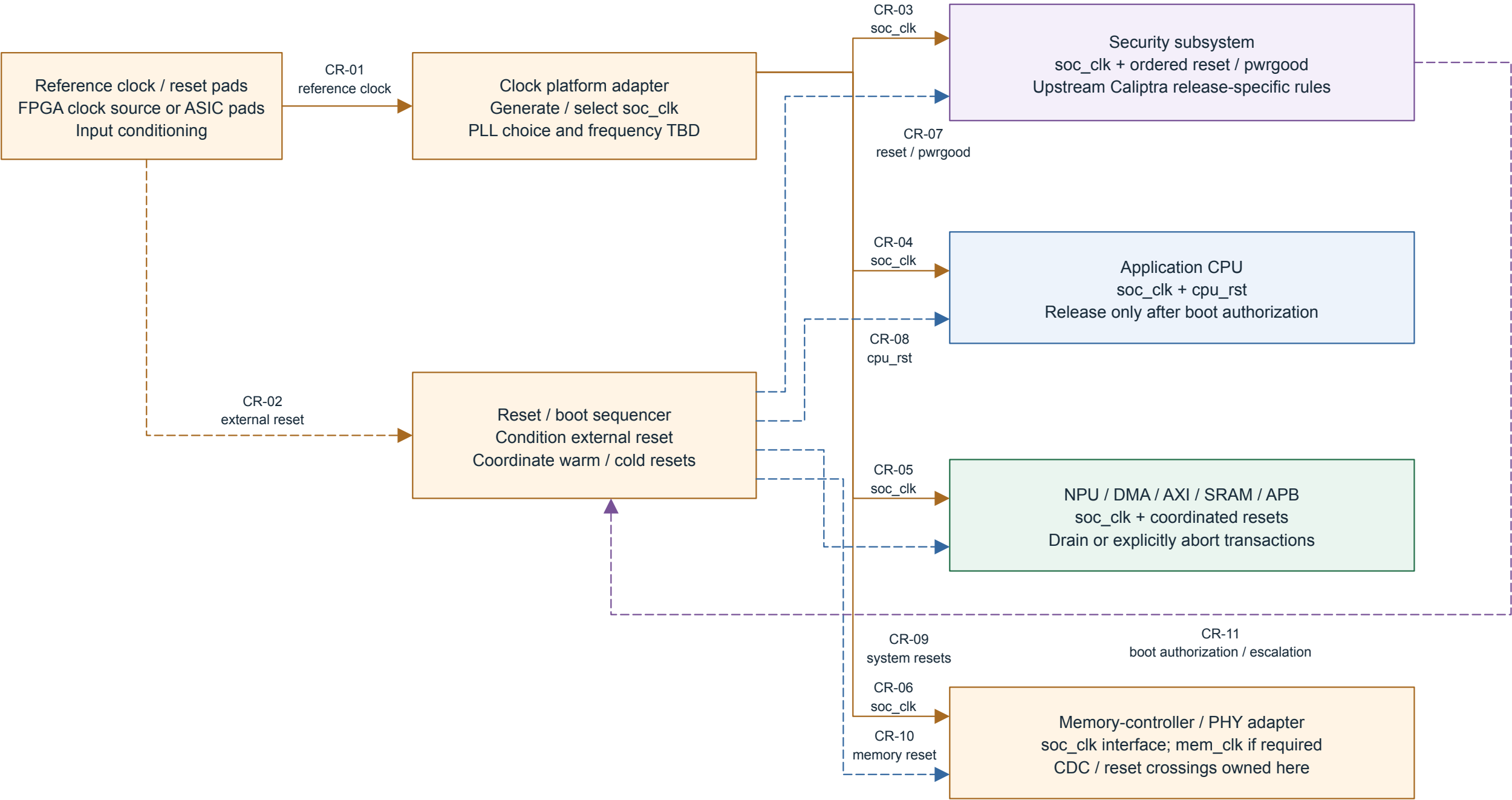


Clock/reset implementation view: control, NPU and security are logical domains; independent clocks / power islands are not yet selected.

Functional bring-up is a separate non-security acceptance stage. FPGA OTP/entropy models do not prove production silicon security.

OpenChip | Clock and reset connectivity

OpenChip architecture review | Rev 0.2 | v0 candidate: one functional SoC clock; external-memory PHY is platform-dependent



Clock lines share `soc_clk`; reset branches are separately controlled outputs. Physical clock-tree synthesis is not represented.

No independent power islands selected in v0. Low-power states, isolation, retention and DFT controls require separate implementation views.